

ActInf GuestStream 071.1 ~ The empirical status of predictive coding and a

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Hello and welcome. This is Active Inference Guest Stream 71.1 on February 22nd, 2024, and we're here with Ryan Smith, Rowan Hodson, and Mariska Mehta. There will be a quick overview and then discussion on their recent work, The Empirical Status of Predictive Coding in Active Inference. So thank you all and let's hear about it. Okay, so you just want me to jump in and go? Yeah. Okay, cool. Well, thanks a ton, Daniel, for inviting us on. It's fun to get to present some of this stuff and hopefully get it out to the broader community a little bit. So I'm going to quickly just kind of walk through sections of the paper to orient people to generally what the point is and what we're trying to do. And then at that point, I think the goal is just to kind of launch into discussion and see if we can kind of extract out some of the more interesting points that might be most relevant or interesting to the Active Inference community. So just to start out with here, so the paper, as Daniel said, is called The Empirical Status of Predictive Coding in Active Inference. The main point of this is just that, you know, as a lot of people in this community know, it's, you know, at its kind of beginning and for a long time, you know, like the Active Inference is primarily, so the literature you'll see has been very focused on theoretical sort of conceptual work and simulation-based work, right? So you'll see a lot of things out there that's kind of, you know, an Active Inference model of X, where X is just some interesting psychological phenomena or condition. And usually that involves showing some, you know, fun, interesting simulations that are kind of potential computational explanations for, you know, whatever the, you know, specific phenomenon of interest is. And that work is great, and I think there's been a lot of developments there, but, you know, at a certain point, you know, we can come up with as many theories as we want, but without actually being able to test them scientifically, it's really hard to be able to say with any confidence that these are kind of accurate stories of what the brain's doing, and what are the kind of, yeah, again, like empirically supported theory, you know, what, which of all these different sort of models, the simulations people are proposing sort of actually correspond to what the brain's doing, and whether they can actually explain human behavior. So what we were interested in doing is actually kind of taking a step back and looking at what the, what the empirical studies that have been done using these sorts of modeling approaches, what they actually say, and how supportive the, the evidence actually is for, you know, the hypothesis that the brain is doing things like predictive coding and Active Inference. So that's why it's called the empirical status, right, is we're trying to say, okay, what is the current evidence, and where is evidence missing, right? So what should future work and future studies focus on to, to try to actually, you know, like, like fill in and answer questions and provide additional support, or not for, you know, for, for these theories as hypotheses for what, for what the brain is doing. So to kind of walk you through sections here, so we focused, you know, so people use this umbrella term of like predictive processing, but that's really a fairly

vague overarching kind of umbrella term. And so we picked kind of the primary, you know, actually sort of like well-defined algorithms, you know, in, that are most prominent within the, kind of under the predictive processing umbrella, which is predictive coding and active inferences, a kind of well-defined mathematical algorithms that can be sufficiently precise to test empirically. Okay. So the first section is more or less just kind of an introduction that says more or less the sort of thing I just said about predictive coding and active inference being the kind of most prominent, precisely formulated algorithms that the brain might be doing, and the importance of testing these things empirically, et cetera, et cetera. And so I should say that, you know, most of the credit for this paper should really go to the first author, which is Rowan Hudson here, who I'm hoping will certainly say much more after I give this kind of brief overview. But, you know, so what Rowan did is he started out in each of these sections by defining the algorithm and laying out the associated mathematics in a way that was really very clear, I thought. Of course, I'm bragging a little bit about a paper that I'm on. So take that with a grain of salt. But I thought he did a very good job of laying it out, you know, being explicit, showing the mathematics, but describing it in a way that was what I think should be very accessible to people that don't have a ton of mathematical background. And the idea was that we would just, you know, introduce this so that what we were talking about was clear. But the focus really isn't on, you know, just the mathematics. We're just introducing a mathematics so it can be kind of more understandable what, you know, what the studies that we review are actually testing. And so we just kind of walk through a lot of that. And I'll come back to this figure in a second. But, you know, the underlying mathematics of, for instance, you know, here is a definition of the negative free energy when applied within predictive coding and just kind of how you get to some of the predictions and prediction errors and things like that that we actually use. And this figure is a figure that took a long time to make and to try to be clear on. But, you know, we thought it ended up being pretty, you know, colorful and engaging. And it looks complicated, but we tried pretty hard to kind of walk people through step by step with this is what this is hypothesizing. But this is just a representation of one way that cortical columns in the brain might implement hierarchical predictive coding. And this is a particular form of a particular predictive coding algorithm that allows for a certain type of temporal depth to it, where you have essentially a higher level hidden cause that's being represented that actually predicts dynamics in the representations at the level below, as opposed to just predicting and trying to minimize prediction error with respect to some sort of static equivalent temporal scale representation at the level below. So we try to kind of lay that out so that people have an understanding of what, again, what the kind of general basis and what the hypothesis is. It's kind of important here because predictive coding at an algorithmic level is a different thing to test than trying to test a theory about the specific neural mechanisms that implement predictive coding, right? So I think it's a really important point is that you can test a theory empirically by looking at the brain to try to, like, test a hypothesis about a specific way the brain could be set up to do predictive coding. But the brain could be set up in more than one way that would implement predictive coding.

So to be specific about testing something about the brain process, you need to specify what algorithm you want to see whether there's evidence that the brain is doing that. More generally, though, you could also just test for evidence of predictive coding at the algorithmic level. And you could do that behaviorally just by looking at whether, you know, people's, people's, what people detect perceptually and how that evolves over time, whether that is just consistent with the, with the predictive coding algorithm. So just that, just the mathematics, not some more specific hypothesis about the way the brain is doing it through different patterns of connectivity. So then, as I mentioned, we kind of go through now, just we lay out what they propose neural implementation is, or as I said, one commonly proposed implementation. There's actually a couple that we talk about. And then here in this section, we have the empirical studies of predictive coding, and we review more or less what the evidence is, for these different, for, for different kind of aspects or testing different predictions that predictive coding might make about what you would see, either behaviorally or, or in the brain. And I won't kind of go into it in detail here, we can obviously talk about it when we get more into the discussion. But, you know, take home is, is really, there is a lot of indirect evidence. But more, there's a lot of studies that still kind of remain to be done to test more specific hypotheses about the neural implementation. You know, whether or not, for example, there actually are separate neurons in the brain that are responding just to prediction errors, and other ones that are representing just predictions and things like that. There really is more kind of explicit model fitting that needs to be done to test out this, you know, predictions associated with really, well, quantitative simulations, and what those predict, based on the predictive coding algorithm. One place where probably the most kind of related evidence is, is not with predictive coding proper, but with a hierarchical Gaussian filter, which is a different model than predictive coding. But it, it has related, it has, it predicts dynamics that are related to predictive coding. So for instance, it does have prediction errors in it, and it does have a precision weighting on those predictions. And it can be, the updates can be modeled in relation to prediction errors. And one difference, though, is, instead of being hierarchical, in the sense of each level in a hierarchy representing different causes, in the hierarchical Gaussian filter, higher levels represent predictions about the stability of the contingencies at the level below. So, so basically, the highest level is representing something like, how quickly it expects the predictive relationships between hidden states and outcomes, how quickly it expects those will change over time. And so it can do this kind of like dynamic precision weighting, depending on essentially how much you should trust the predictions you have in a given moment, based on how you think those predictive relationships will change over time. So it's not the same thing as predictive coding, but it's related. And people have done neuroimaging studies, for example, and looked at the explicit relationship between the prediction error dynamics that this model, the predictions it makes about those prediction and prediction error dynamics, and whether or not you can find neural responses that look like they match those predicted simulated time courses. And there haven't been a few studies that do support that specific brain regions are encoding prediction errors and predictions of things like that, and different precisions. So it's not the

same thing as testing predictive coding directly, but it is testing and finding evidence for a neural basis of encoding of precision-weighted prediction errors, which again overlaps in interesting ways, as kind of again,

indirect but important evidence. So then after we go through that, then we switch to talking about active inference.

And again, kind of similar structure. First, Rowan introduced the mathematics associated with the active inference

framework. And, you know, for people that are watching this, you know, this, you know, the active inference institute program regularly, should be aware that active inference, you know, nowadays has evolved into kind of a, also being a broader umbrella

term for multiple different specific mathematical formalisms. And so to be clear, this, this one that we go through here is, is just the kind of standard

initial partially observable markov decision process framework. So this is a specific model of decision making based on a particular explore exploit

trade off in a model based way, where instead of just using reward, as a cost function, or objective function, it's based on the expected free energy, which is just this combination of expected reward and expected information gain.

So it's a, it's a, talking about a specific algorithm and kind of the original algorithm under active inference, but now there's, there's lots of different kind of variants of it. So we're just going through the basic one.

So again, you know, we talk about how it's related to predictive coding, and what's different about it from predictive coding. And then in this case, what we do is there is actually quite a bit more, you know, quite a bit more in active inference, at least this version of it, there are a number of empirical studies that have directly fit active inference models to decision making behavior.

You know, several of these come out of our lab, actually, but there are some other ones that haven't. And so we reviewed each of these in detail, kind of laid out how the model, how the models were set up.

So we're set up to model specific decision making or, or perceptual decision making tasks. You know, so for instance, here, we kind of introduced the model for a specific task we use to model interoception, so cardiac interoception.

And then yeah, here, you can just see this is just kind of a standard, you know, version of a depiction of the active inference model.

And then let's see, you know, again, these are sort of standard things you see in lots of active inference papers, right? So just describing and laying out the mathematics of the of the general update equations and the overall algorithm and then

a very kind of coarse grained generic sort of representation of one way that the brain might implement those algorithms kind of inspired by cortical column structure.

But so anyway, you can see this is a model that we use to an active inference model we set up to model a three arm bandit task.

that we used in in substance use disorders to show that and in this one, you know, what we were able to show, for example, was that substance users showed slower learning rates from negative outcomes than healthy individuals.

So you can see it's much more based on trying to answer psychiatric questions about clinical groups because

our lab is focused on computational psychiatry in particular.

And to walk through that, we walked through a task that we modeled using an approach avoidance conflict paradigm.

And that model also was able to show differences in decision uncertainty and what we call the emotion conflict.

It's a type of preference precision effectively that also differed in affective disorders and substance use disorders from healthy controls in an interesting way.

And there was also a couple others, one really cool study that was not by us, but by Gibson was the first author.

And they took they took, for example, a set of publicly available data sets for what's called a two step task, which is just a common task used in reinforcement learning models.

And in that task, there's a standard reinforcement learning model that's used that's essentially kind of a mixture based model of of what model free reinforcement learning model versus a model based reinforcement learning model would predict.

And what that task was designed to do initially was to use this kind of mixture reinforcement learning model to test individual differences in how kind of model based versus model free people are.

If we can talk about that a little more if people, you know, listening, aren't totally familiar with model based versus model free distinction.

But but what they did, which was really which was really cool was they took that data and fit an active inference model to it and then compared the ability of active inference to explain behavior compared to the standard reinforcement learning model I just mentioned.

And what they were able to show is that the active inference model in two of the four data sets was actually a little bit better at explaining behavioral patterns than the reinforcement learning model was in the other two data sets.

The models were about equivalent.

So that was probably the most direct evidence that that has been shown.

And this was one of the major points I think that that we wanted to make is that even though it's the case that, you know, us and other people have started to do actual empirical studies trying to fit inference models to data.

The main purpose so far and what people have done really hasn't been to provide unique evidence for active inference.

It's just been using active inference as a way to try to identify individual differences or group differences and in a psychiatric research context.

So so so one thing that is still really needed aside from this one kitchen study I mentioned is more studies where people actually take, you know, behavioral data or collect behavioral data on decision making tasks and fit a bunch of fit different active inference models to it and also fit a bunch of sort of competing models like reinforcement learning models to it.

And to really show that the active inference that the behavioral patterns that active inference predicts are that active inference can can do a better job of predicting that behavior than competing models can because that's really the only way.

It's not enough basically to show that active inference models fit well.

You need to show that active inference models fit better than competing models because that's how you'd really show that that active inference is more well.

You need to show that active inference is more likely what the brain is doing when people are making decisions.

And so, you know, the overarching conclusion is that, you know, the evidence is promising, right?

I mean, there's nothing that suggests that the brain isn't doing active inference.

And certainly it's consistent with people doing active people using active inference, but a lot more still needs to be done to show that active inference is a better explanation for what people do than simpler competing models.

So that's, you know, the way that the brain is doing is a better understanding of what the brain is doing.

So that's, you know, just as a kind of broad brief overview of the general structure of the paper and the message we're trying to get across.

So, you know, we can we can certainly talk about it more just kind of interactively now.

There's lots that I didn't talk about, but hopefully that starts as just kind of an initial launching point for discussion.

So I guess I'll just stop there.

Thank you.

Awesome.

Perhaps the other authors could give the first takes introduce themselves.

Sure, I'll start.

Yeah, I'm I'm Rowan.

I'm a PhD students.

Ryan's my supervisor.

I'm at the Roy Institute for Brain Research.

I did my Masters at the University of Cape Town under Jonathan Schock, Mark Soames and Ryan as well.

I think, yeah, just to talk a bit more generally about this paper, this paper started off as a book chapter, actually, which was only going to be focused on active inference and sort of especially particularly parallel states of active inference.

In the process of writing this, I found myself what needing almost to write about predictive coding.

It's it's I think whenever we can, we should take the opportunity to really present this in a in a very methodical and complete way.

And I think sometimes I remember when I was first learning about active inference, it's very hard to dive in in them into the middle.

And I think that's how it sort of became it more from just being focused on active inference to being focused on active inference and predictive coding where you print of coding access a sort of foundation for a large aspect of active inference.

And I think that's how that sort of happens.

And then, of course, it morphed a little bit from just just talking about predictive studies to also talking about the sort of the background mathematics and theoretical foundations.

And so I think this is the sort of field where it's a noble cause to try whenever we can get the opportunity to try

present a quite a thorough review of all aspects of this because it's a difficult field to learn.

It can be very confusing. So, yeah, I really hope that this review I'm on top of presenting the empirical side of things also acts as a reasonably intuitive way to look at the foundational theory of predictive coding.

I think it's a very important thing.

Yeah. And one thing I will say just to kind of add to that is, is that, you know, for people that, you know, have more of a kind of broad conceptual or philosophical interest in active inference as opposed to, you know, more of the kind of like detailed mathematical modeling sort of understanding.

Often what I've seen in the past is, is that predictive coding and active inference get a little bit kind of conflated.

Right. So, you know, people think that active inference is somehow just predictive coding plus motor control or plus decision making or something.

And so making a claim that's actually really not the case, I think is important. Right.

That predictive coding is actually an entirely different algorithm. It has a different generative model associated with it.

You know, it uses continuous state spaces, whereas active inference uses discrete state spaces and discrete time.

And so, you know, I think Rowan did very well, you know, with Mariska's help also, was, you know, to kind of state clearly what the connection is between predictive coding and active inference, but that they are not the same thing and they're not even directly connected to each other.

So to make that to make that to make that a little bit more more clear and more intuitive, maybe.

So that was just another thing I wanted to say that I thought I thought that they that they did pretty well and is another kind of maybe useful take home point from the paper.

Hi, I'm Mariska. So I'm also like Rowan, a second year PhD student at the Laureate Institute and Ryan as my supervisor.

My background is more neuroscience and psychology.

So this paper was like a really cool and challenging first step in my PhD, where I come from a more empirical background.

So kind of working on this paper was kind of going not taking baby step, but deep dive into the active inference and the predictive coding literature.

And again, I'm the people that Ryan was talking about who had more interest on the empirical side and kind of going through this journey, like really helped me like walk step by step into like different aspects of the just like predictive coding or active inference literature that you should take into consideration and think about really like the algorithmic side, the neural side of things and kind of like connecting those three levels, how we how different people

kind of just think about from one perspective, but what really is needed is connecting all those three things.

And I felt like I was playing a supportive role in this paper, but this really was a great learning journey for me.

Okay, why not to Mr. Empirical Dean.

So my interest in this was anytime I see a title where there's a minimum of two ideas being brought together as the unit of analysis. I'm always curious.

Is it okay if I just because I have one question that's sort of an unpacking question around predictive coding and one question related to active inference at the conclusion would it be okay if I read a little section of the

paper?

Okay.

Sorry, just start just there's a little microphone thing.

Can you just repeat again?

Okay, so I'm on page eight of the paper.

And the section that I like to get a little bit of unpacking around is, thus fitting predictive coding models to responses on perception tasks and testing or quantitative predictions from simulations remains an important direction for work going forward.

This is crucial because simulated dynamics and predictive coding models can make predictions that are not always straightforward a priori.

And that depend, like this, depend on the specific hypotheses built into a formal model.

Ryan kind of touched on that in his introduction.

For example, the specific mathematical form assumed for the mapping between levels in a hierarchy, the direction in which neural activity is assumed to represent a particular posterior estimate.

So often, this is the thing I was hoping somebody could unpack for me, often sequential dependencies between task trials and patterns and dynamics can be missed in summary statistics that average over trials.

So models are necessary to predict and test for the presence of those precise dynamics.

I was hoping maybe somebody could put some color on that, that process and what it means.

Sure, I can take a stab at that.

And I mean, you know, Rowan or Mariska can as well, but I'm pretty sure that was something that I wrote.

So maybe that's for me to be the one to at least take a start on it.

So, yeah, I mean, in the field of computational psychiatry more broadly, right, so not just active inference, but, you know, the goal is really just to, you know,

really just to take a set of task behavior, right?

So we can ask people to do, say, some kind of like sequential decision making tasks that involves usually like explore-exploit dynamics of some form, right?

So say like one common one would be I give people like a three-armed bandit task, right?

So I give people that play some little game and there's three different options and they don't know what the reward probabilities are.

So, and then they can just trial and error in the beginning.

They can say choose option three and they can see whether they win or lose.

And if they win, you know, let's say they probably stick with option three again.

But if they lose, maybe they try option two and then say if they lose again, they could try option three again or they could try option one and so on and so forth.

Right? And so predictive and active inference model in this case, I'm just using this as a kind of a simpler starting example.

An active inference model under different parameter settings in that model, right?

Under different learning rates or under different directed exploration drives or however you want to parameterize it, would make different predictions about what that sequence of choices would be.

And there's a sequential dependence because the choices on each trial are not independent, right?

So what a person chooses on trial two depends very much on the outcome they saw at trial one.

And what they see on option three, what they choose on option three will depend on both what they saw, you know, after choice one and choice two.

So if I do this standard, right, empirical thing that you would like analytic thing that you would do for a task like that, that's not model based, is you might just have some summary statistics, right?

You might do something like a common thing to do would just be to count the number of times they switched what option they chose after they lost, or how many times they stuck with the same choice after they lost.

So we call those like win stays or win switches.

Or same thing, like if you lose, did you stick with the same one or did you switch?

So it is win stay, win shift, lose stay, lose shift.

You could just kind of count the proportions of those, right?

So you could see something that, you know, maybe, you know, how often people lose switch, for example, might tell you something kind of like their learning rate for losses, right?

Like they, they might learn more quicker, they might learn more quickly, right?

Update their beliefs more after a loss.

If they, if they're the kind of person that switches away, right, they'd be more quick to assume, okay, if this option led to a loss this time, that's probably not a good option anymore.

So I'm going to switch to something else, right?

Whereas if a person has a slow learning rate for losses, then maybe they'll stick to the same one a couple times before they decide, okay, this is definitely a bad option.

They have to see like a few losses in a row, right?

So, but the point is, is that if all you do is kind of average over trials and get some kind of summary statistic about how many times people, you know, win shifted or lose stayed, etc.

What you're not going to get is anything about the pattern in the dynamics, right?

So if they, like, for instance, if they like lose shifted in the early first few choices, right?

Like that tells you something pretty different than if they lose stay or lose shift on some of those later choices, right?

The early choices might be driven much more by like exploratory drives, right?

By information seeking.

Whereas on later trials, that much might be much better explained by differences in learning rates.

So, so fitting models to behavior, which again, just means you have the actual behavior and then you see what the model predicts under a bunch of different parameter values, right?

And you try to find the parameter values for a person that best reproduces their behavior, right?

So point is, is that we've done that kind of thing several times now for active inference, right?

Which you can do with just by taking patterns of decisions, right?

That people make on games.

But you can do that same thing for predictive coding, right?

You can give people perceptual decision making tasks, right?

Where people say, okay, this is what I perceive this time.

Okay, now this is what I perceive this time, etc.

And those are also going to have sequential dependencies because people are going to build up prior expectations about what they're going to see next.

Right?

And so predictive coding models, again, under certain parameter values might say, okay, well, if they saw the same thing the last five times, they're going to probably a lot more biased, right?

They're probably going to have a much more precise prior that they're going to see that again.

So the probability they're going to say they saw that again, even if you showed something different would be higher.

Right?

And so there are, the point is, is that it's not always the case that a predictive coding model is going to predict some trajectory of choices that's going to be really obvious without actually fitting the model.

You know, if, for instance, the person, like, has no tendency whatsoever, to be more likely to say they saw one thing just because they saw it a bunch more times in the past, right?

That wouldn't really be very consistent with the idea that people are using predictive coding.

And it gets even a little trickier than that, because predictive coding is based on continuous state space.

So it's not really even something like a person would be, you know, just like choosing, I saw this, I saw that.

It'd be much more something like them kind of continuously turning a dial or something, you know, as they see something get brighter or dimmer, or, you know, like motion in a direction or another direction.

And because the prediction error equations are set up in this kind of continuous way, they also have these kind of like oscillatory dynamics to them.

Right?

So it's like, not like you get error, and then it drops as the thing resolves, but it kind of oscillates up and down a little bit.

Right?

And so that would also predict pretty kind of sometimes funny, like not, definitely not something you could just predict a priori, like exactly how someone's going to like turn a dial.

So that's, that's the kind of thing I mean.

Okay.

In my in very simple terms, Daniel, I've had some conversations around, if you're in a, if you're in a situation where your focus or your concentration is on the next move, or if you can look and see the entire space and take an all moves perspective.

So my asking you the question was to get out of the sort of model piece of it and that, so what what do people actually do and I think you're the question around the oscillate there that's around the oscillation.

Sort of speaks to that it's, it's difficult when you're focusing on that next part of the sequence to be able to take in the entirety of the

And vice versa if you're focusing on the averages, maybe you're not able to pick up on the nuances of the next step.

Can I ask one more question?

Sure.

Or would the other authors like to continue to fill in on that first answer?

I think Ryan explained that very well.

I think this is a general, I'll talk about it, I know we that is referring to a predictive coding, but in something like it's something I'm very interested in active inference is along the lines of differentiating, for example, between a model that uses reinforcement learning and a model that uses active inference.

And yeah, using summary statistics sometimes it can be hard to actually see differential behavior, right?

This is what we care about when we are looking at comparing different models.

What do they actually predict differential behavior?

And that differential behavior can be difficult to capture in summary statistics sometimes.

So in general, I think in the world of sort of model fitting, this is just a common common theme.

And as Ryan said, while we're doing this active reference, they are limited.

There was a lot of applications of that methodology in predictive coding.

So.

Okay, Marish gives us a lot.

So I'll ask this, the other thing, this was in the conclusion of the paper.

And again, I'll just read it.

In contrast to predictive coding research, in contrast, predictive coding research can be traced back nearly four decades and make specific predictions that can be investigated across a variety of fields.

So there's a deeper well of priors there.

I assume that's what that means.

It will be an important direction for future research to further develop the neural process theory underlying active inference and allow for precise implementation level, as opposed to simply algorithmic level predictions about random behavior.

I completely agree.

Until that time, confidence and active inference as a neural model of decision making should remain tentative.

But this is the part that I thought was really interesting.

He wrote, another important limitation associated with the current active inference scheme is that of scalability, which constrains the phenomena that can be examined in empirical studies.

That is, while these models work well in the context of simple tasks, become less tractable if applied to many real world problems, side dimension sets of states, observations and policies.

My question was, is the idea of strategy because it's active inference is known kind of as a basis for strategizing behavior.

Is the question around scalability one that ties into the idea that even when humans are trying to strategize, how far out they can generalize their strategy is a difficult measure to get precise, given that most contexts are dynamic and changing.

So there's several kind of related things here.

So, so the, the tractability issue kind of comes in two different flavors.

So one flavor is tractability with respect to like our, that our ability to like use these models to even simulate

behavior in, in, in contexts where the, the kind of space of options and how far in the future people are planning.

Right.

When that gets big.

Right.

The other, the other question is more about psychological plausibility, which is that, um, even if I can get, you know, an active inference model to simulate kind of like 10 steps ahead.

And you know, where there's like, ends up being like 30 or 40 different combinations of like 10 moves or something like that.

Right.

Um, it's not really very plausible that humans are really doing that.

Right.

That the brain is really doing that and yeah, standard computer.

It's going to take like, I dunno, like Ron, you know, Ron has these, uh, fun planning models that he has this paper that, um, you know, we, uh, I think are pretty close to submitting.

There's a free print of an earlier version that's out, um, where, you know, the thing does like plan, right.

A bunch of different possible paths that it can take to, you know, find, you know, different sorts of rewards.

Um, and, uh, you know, it takes like hours, right.

To run like simulations on a computer to do something that however humans are doing it, they can do it just like, you know, in a minute or something.

Right.

So, so, um, you know, I don't, so it's just not very plausible that the way that active inference is solving problems like that, um, is in any kind of exact form, the same way that humans are doing it.

Um, so there's kind of scalability with respect to what humans can do because humans can't do this explicitly in a fully model based way.

It doesn't seem like, um, after a certain sort of level of complexity to the planning problem that needs to be solved.

And then the other part is just tractability with, you know, actually doing the modeling itself, uh, because of just like how just, um, intractably long it could take, um, to actually run these simulations.

Um, and so both, both of those things kind of come into play and it's not super clear exactly how, you know, to address this.

I mean, there's, there's certainly ideas out there, but most of them involve, um, taking additional kind of machine learning tricks.

Um, you know, like adding, like adding like deep neural networks to active inference models, for example, or, you know, doing, doing, um, you know, various other sorts of, um, you know, little kind of heuristic short cutty things that make stuff tractable, right?

Like not actually searching all the way down every possible branch of a decision tree, but using some heuristic to kind of say two steps in or something.

Nope. This branch seems bad. I'm just going to cut this off and not consider it anymore.

You know, things, things like that, or like sampling based approaches, um, for exploring just kind of little bits of the decision tree at a time.

Um, you know, it doesn't, it's no longer pure active inference anymore.

Right. I mean, it's, it's a kind of combination of active inference and a bunch of other machine learning things.

Um, but you know, to a certain extent, like that's probably just like necessary.

And I mean, bottom line is the brain probably has to be doing something other than fully model based active inference, um, to be, to be tractable.

And so figuring out exactly what parts of what the brain do is doing might be pure kind of active inference and what parts are these other things to make stuff tractable, um, is, is I think part of the question.

Yeah. Look, the scalability issue with active inference is not an active inference issue. It's just a Bayesian learning and decision-making issue.

Yeah. Um, it's, there's nothing inherent to active inference that is, that slows things down.

Um, any, any technique that can be used in any other sort of Bayesian decision-making method can be used in active inference.

Um, so yeah, this is a, the grand question of how the brain is so efficiently able to do this, um, where, you know,

where there's, yeah, like Ryan said, I mean, you run these very basic tasks and because of this, the exploding state space of, uh, Bayesian decision-making, right.

Where does this, this is exploding state space of probable probabilities, um, you get a massively expanding search tree ultimately.

Um, and yeah, nothing to do with active inference. Of course, active inference is, um, a Bayesian framework.

And so it's, it has to sort of use this. Um, but that's just, uh, I mean, if we can solve this, then, uh, that's, uh, yeah, it would solve a lot of things.

If we could solve how, how we can perform Bayesian inference and especially Bayesian inference in the service of decision-making, how we can do that very efficiently.

Um, that would solve a lot of things. Um, but as of yet, yeah. If we want to try speed active inference up, we have to speed the general field of Bayesian decision-making up.

Yeah. Yeah. Oh, it's good. Sorry. Yeah.

Okay. I was just gonna say is part of it, perhaps from a strategy standpoint, not trying to generate the world's biggest plan, but actually going back to that rules business and saying,

can we start there instead of a plan that because of its, it's just its size makes things intractable. Can we, could we swap something else in for something we know won't work, which is world's biggest plan?

Um, I mean, I mean, look, there's lots of different, you know, things that, you know, might, you might consider like, like one, one thing, you know, for these, like,

exploding decision tree, um, sorts of problems is, you know, finding some way to, um, chunk things together, right. To make like the, you know, so for instance, instead of, you know, say I'm going to go, I decide I'm going to go walk to the store or something.

Right. Like in a certain sense, I've got a ton of different options about, you know, where to put my feet at each step and, you know, like what door to go out of and things like that.

Um, but it's not really clear that when I'm planning, like to go to the store that I'm really explicitly considering

all of those details, right.

Like I can chunk it into, you know, I'm going to walk out to the street, you know, I'm going to walk, you know, 10, you know, I'm going to walk a mile to the store and then I'm going to walk into the store.

Right. In which case, if I've chunked that to just kind of three steps, right. Then my decision tree is, is already much more tractable.

And then you just need to tell some story about, you know, how, you know, when we get to certain points and that really abstract chunked, um, plan, when we get into those kind of chunk states, how we make these kind of more local decisions about what to do when we're at the, at smaller scales, like when we're in those states.

So, you know, there's lots of kind of hierarchical chunking ish sorts of things that, you know, you might think that, um, the brain could be doing to make these things tractable, but, and there's lots of different kinds of options on the table and things that people, um, might try, but all of things have certainly, all these things have not been, um, kind of, you know, tested against behavior to see, you know, which ones are, are, are the most plausible, but there's certainly, there's certainly different possibilities for what might be going on.

Um, you know, another example, you know, this is something that we test empirically in my lab right now is, um, the way people might do something called like aversive pruning.

Um, and so, so what you're kind of doing there is you're just, and this is kind of similar to what I said before is, you know, when you start kind of planning down some tree, if there's some early outcome where you're imagining that it's going to lead to some really negative outcome, right?

Like on like, if like on step one or step two and a possible plan, I think, oh, there's going to be some big negative thing.

Then I will just no longer simulate down the rest of that branch.

So it's a way of kind of reducing the number of branches in a tree after search.

And, um, and so that's, it's often called again, aversive, aversive decision tree pruning.

And that, um, you know, obviously you need to do something like that, right.

To keep it tractable, but at the same time it can cause problems, right?

Cause sometimes the best plan might go through something negative in the short term, but lead to the best thing in the longterm.

Right.

I mean, so that's the kind of thing we, we test in the lab is whether or not different psychiatric conditions involve kind of doing pruning too much, right.

Or too little, um, and how that could lead to some optimal behavior in the longterm.

Um, the other, I think the other aspect of the previous question you asked also had to do with, you know, you read part of the thing talking about, um, the, uh, the, the neural basis of, of active inference and how that should.

Kind of remain tentative.

And, um, just to kind of clarify that a little bit, I mean, the, I mean, the main point is, is just that, again, this kind of involves being clear about the separation between algorithm and implementation.

Right.

So the, the algorithm is basically just the mathematics, right.

That we, um, you know, that we kind of lay out, but then the implementation question is, you know, what are

the different possible ways you can kind of set up the brain to, to carry out those equations.

And it's a separate question.

What the, you know, how the brain, um, how the brain is doing that, even if you think that there's good reason to believe that the brain is doing the, doing something that's well characterized by the, the mathematics.

Um, and, um, unlike predictive coding, which has been around a long time and, you know, there's been much more opportunity to test, um, at least, um, qualitative predictions.

Right.

You know, things like whether or not there's evidence for like omission responses in the brain, right.

Like when there's the lack of a stimulus, when it was expected leads to an oral response, um, which is, you know, one of the stronger pieces of evidence that the brain is doing something predictive, like predictive coding.

Um, you know, for, for active inference, uh, there really has been like one imaging study and it was done like in 2015 and it was like an older version of active inference.

It wasn't the current full based on the current formalism.

Um, you know, so there's, there's just, it just hasn't been done.

Right.

And the, and even the, um, you know, the little kind of column structure things like what's in the, um, like what's in the review in our, in our, in this paper, um, is super just kind of like promissory heuristic sort of thing. It's just like, here's a bunch of, here's a bunch of little balls that will pretend are like neurons.

And here's how you could connect them together roughly, you know, to, um, do some sort of, you know, some message passing algorithm that, you know, you're thinking is, you know, you're imagining or hypothesizing is the, is the one that the brain's doing for this kind of approximate inference process.

Um, and, um, you know, even when it comes to, um, the current mathematical proposal, um, for how the brain might be doing something like this, um, there's different hypotheses, right?

So like initially most people, when they were doing simulations with active inference models were assuming, um, something called variational message passing, which is just a particular way of kind of repeatedly doing local approximate Bayesian inference on different kinds of nodes in a graph.

Um, and they, you do this kind of over and over again, the thing kind of converges to a good guess about what the posterior should be overstates at each time point.

Right.

But, but, but, you know, then after that, a little bit, um, more recently, you know, Thomas Parr and Carl, um, and, uh, maybe other people were on the paper.

I can't remember.

They proposed a kind of updated version of that called marginal message passing, um, which is a little better.

Um, it does a little, it's slightly more, it can get at better, um, approximate posteriors, um, you know, on average.

Um, but then there are others, right?

There's belief propagation is another sort of message passing algorithm that's, you know, been considered.

So there's a bunch of these, right?

And each of those, even if they're doing active inference, right?

Like they will also predict different neural dynamics.

Um, so, so those are also even just active inference under what message passing algorithm, um, separates into a bunch of different competing hypotheses, um, about what you would measure in the brain.

Awesome.

I'll read a question from live chat.

Andrew P writes, curious for Dr. Smith, where might he place his previous work, like simulating the computational mechanisms of cognitive and behavioral psychotherapeutic interventions insights from active inference in relation here.

Would this kind of work be more on the side of theoretical exploration, like the cognitive affective behavioral interactions construct, or is there a direction for empirical testing?

I mean, as is absolutely.

It was a theoretical simulation work sort of paper, right?

I mean, we weren't fitting that model to any sort of empirical data.

Um, it does.

I mean, that one, that paper in particular does lend itself to at least certain that, that model is a little too general.

Like the patterns of behavior that it predicts are really not specific enough to like test in a, you know, some sort of task.

Um, you could, you could make it, you could modify it in a way that might make it specific to some sort of, uh, task that would involve, um, you know, specific sorts of explore, exploit decision-making choices under kind of expected negative outcomes, things like that.

Um, but, uh, it does make a kind of sort of qualitative prediction, um, that could be, that could be tested.

Um, you know, so one, one, one parameter that you might fit, right?

Based on that model that I think would be super interesting if you could kind of turn it into a task is this, um, parameter that reflects like the degree to which your cognitive beliefs influence your automatic affective responses or expected

affective responses.

Um, the, um, so, I mean, it would be really interesting to, to try to use that to figure out whether or not there are, we can measure individual differences in this kind of, uh, what, what often gets called like cognitive penetrability.

Right.

So whether or not I, you know, cause I might explicitly believe I'm safe, right.

In a certain situation, but I might see something that still makes me feel really this automatic kind of fear response in my body or something, even though I explicitly believe it's not.

Oh, I explicitly believe it's safe, right.

My affective responses don't always have to match my explicit cognitive thoughts.

That's often a thing you see, right.

In people with, with, um, with affective disorders.

So this kind of individual difference that you could estimate about, uh, the degree to which these two, um, essentially these two, uh, um, um, different, uh, hidden state factors, you know, in the, in the model, the degree to which those things interact effectively.

Um, but the, the other, the other, um, prediction that makes, um, again, it's a little more qualitative, but certainly something that could be tested or made more precise is, um, that, you know, in that, in that model.

Um, one of the interesting things that came out of the simulations is that, um, you actually, it's probably actually not a good idea to make people think that they're explicitly believe that they're in a safe context.

Um, before you do something like exposure therapy, um, because if you do that, then basically what you're learning, what you, the, the beliefs that you're updating in your, your likelihood, right.

About like what actions are going to lead to what outcomes, those will be updated under the belief of safe context.

Right.

So that means all you have to do is switch back to believing you're in the dangerous context, and then you're just right back to the problematic avoidance behavior again.

Right.

Um, and there's a bunch of other, there's recent empirical work that's very consistent with this idea actually about, you know, explaining using this kind of latent causal inference, um, framework, not, not an active inference in particular, but just latent causal inference more generally that, um, you know, to really get exposure therapy to work long-term, um, you need to kind of prevent that people are inferring that there's a new latent cause in operation when people are doing exposure.

Um, cause otherwise they're, they're just learning safety under, under a context where they can really easily get the, get their, um, maladaptive avoidance to come back just spontaneously.

Um, just by kind of being uncertain about the context they're in later.

Right.

Like they're not, they're not actually unlearning the problematic belief.

They're just inferring there's a new cause where under that new cause it's safe.

Um, so, so it was much better, um, the, the, the prediction.

That, that prediction of that, of that paper would, would be, that could be tested.

Right.

And again, like I said, there is now some evidence consistent with this, um, is that, uh, it's kind of better to do things to keep people uncertain or keep people believing that they're, they're still in the same context.

So that when they go through this exposure therapy and get unexpected outcomes, they're actually kind of overwriting their previous beliefs.

Um, as opposed to just inferring they're in a new context.

Awesome.

All right.

I'll, I'll ask a kind of general question that I'd love to hear everyone's perspective on.

Um, we have active inference or predictive coding or some other type of formal framework that does interface with empirical data.

Like we're discussing.

So on one hand we have the empirical system or the measurements coming from it, existing datasets, datasets we create and so on.

And then on the other side of this kind of statistical interface is like numbers, free energy principle, category theory, stuff that is really not having its validity or truth.

Um, or essence described by any particular system.

So how do you just approach this usefully as a graduate student researcher, or as like a broader clinical research program and like, think about what system are you choosing?

What statistical interface and how detailed do you go there?

And how do you think about like what's on the other side of the statistical interface?

Because a lot of times the discussion around like the validity of active inference and free energy principle is a super philosophical question as if a philosophical discussion would resolve uncertainty about the empirical status of active inference.

And what's on the interface?

Which is actually the direction that you all took it from the interface back to the system, rather than interface as outcome from something theoretical to debate.

Um, I feel like I have been kind of hogging the show here a little so I'll let other people answer that first at least.

Um, okay, sorry, that was quite a long question. So I'm sorry if I missed some of it. Um, and just correct me if I'm not answering. Right.

Uh, I think I will talk on this idea between yeah, on the one side, sort of mathematical formulations representing something like brain processes, and then you have sort of mathematical formulations brain process and behavior.

Um, and, um, how these are sort of seems as unified.

I think, um, as Ryan said about, uh, talking about neuronal implementation, um, there's many ways in which, um, we could set up mirror, like a neural, neural implementation to, uh, achieve, um, something like predictive coding.

Or active reference. Uh, the same goes for algorithm where, um, there's many different sort of forms of algorithms could achieve, uh, or statistical windows, as you said, if you want to go more broad, um, could achieve the same, um, neural implementation.

So, uh, and then ultimately, uh, behavior. So, uh, in terms of, and please stop me if I'm not answering correctly.

Um, but yeah, in terms of sort of as a graduate student choosing what to focus on, um, I think it's sort of a evolving process where you constantly just try, read and come up with new ideas about how one can think about how these can connect and how one can represent the other.

Um, and, and for me, it's been particularly interesting just actually understanding what active inference is in its, um, mathematical essence, right?

We've got this first principle account of it, um, divided from the concepts of homeostasis.

And yeah, it's, it's, it can sometimes feel a bit strange when you look at, um, quite simple mathematical implementations in an algorithm and see that, well, this isn't so different from something like Bayesian reinforcement learning with specific directed exploration.

Um, so they are these, there's a sense of, of that sort of first principle accounts, um, that's comes all the way from Markov blankets.

Um, where's that captured in these algorithms?

And, um, that's something I'm still trying to figure out to be honest.

Um, it's, I think it's a, it's in a general, it's a problem, right?

To, and I think that's why, um, when Ryan mentioned sort of, uh, looking at more complex tasks, I think more complex tasks would allow us to explore, um, more complex representations of, of active inference.

Um, that can, that can perhaps capture, um, some of the foundations that Friston laid out.

Um, so does that answer the question sort of?

Sort of.

Yeah.

That's awesome.

Yeah.

I mean, the other, I mean, the other things I would say, I mean, I, I think I understood at least part of the question a little bit differently.

Um, was that, um, you know, there's sort of what, um, I, by that I don't mean that you misunderstood, Ryan, you just answered your different part.

I know, that's all I get.

Just come to my answer.

Um, part, part of what I took your question to be was something about, you know, to what degree answers about the, you know, quote unquote validity of active inference or the free energy principle can be established through kind of like philosophical argument versus what can be, um, or what needs to be.

Um, um, through kind of direct empirical testing.

Um, and, um, and I think, um, you know, here, I think it depends a little bit, right?

So, I mean, I think what, you know, what philosophy is really good at is actually more finding ways that something couldn't be true.

Right.

I mean, I take, you know, I mean, I have a philosophy background, um, as well.

And I think, I think philosophy is really good at like conceptual analysis.

Right.

So trying to find, you know, where some particular conceptual framework, um, you know, where there's tensions, right?

Or where there's things that don't sort of at first, at first need sort of appearance, seem like they're contradictory, have some sort of internal tension.

Um, but when you really drill down, you find certain contradictions, right?

That, you know, rule out something or make it really implausible, um, either through sort of direct logical, um, um, argumentation or, um, through kind of like, um, just proposing sort of thought experiments that really kind of pump intuitions in a way that make a difference.

So, so I think you can rule out with philosophy, philosophical methods, um, things that are conceptually problematic, right?

So you can get down to kind of the set of frameworks or the set of, yeah, the set of sort of interlinked concepts that make up a theory.

You can find a set of them that are at least internally consistent and, and, and have some sort of general sort of sufficient coherence to them.

But once you've weeded things down to the kind of subset of possible theories that are internally consistent, right?

And, and don't have any philosophical problems associated with them, then, then I think, um, things really kind of need to get empirical.

Um, you know, because, because, you know, to the extent that different kind of competing, um, conceptual sort of theoretical frameworks, um, to the extent that they make different predictions, um, and to the extent that the goal of this whole project is to be a different kind of theoretical theoretical frameworks.

Um, and to the extent that the goal of this whole project is to try to come up with an accurate theory of what, you know, what brains are doing, right?

Um, if that is the goal, then the way, the only way ultimately to validate that is by actually looking at brains and seeing what they're actually doing.

Um, and, um, and so, and so I think, I think that's kind of where rubber meets the road a bit, right?

Is, is that, um, um, yeah, you need to know what's plausible, right?

And that's where I think philosophy can help a lot.

But after you know what's plausible, then testing what's kind of actual, right?

Is, is, is kind of the next necessary step.

And, um, and this kind of comes back to what Rowan was saying, um, was that testing the unique predictions of, say, some given active inference model versus some competing model from another framework like reinforcement learning, um, can be tricky.

Um, usually you have to do this in an incremental way.

And you do that by finding, by identifying some prediction that is different between those two models.

And then designing a task that would really put people in a decision, in a, in a position where they have to make choices that are consistent with either the prediction of one or the other.

Um, and, um, it really requires some creativity, right?

And coming up with the right kind of task with the right sort of decision-making demands, um, and constraints, um, that will do that.

Because there's lots of tasks, um, where say the kind of simplest, most straightforward, um, active inference model will really make like almost exactly the same predictions as well.

Like a simpler reinforcement learning model would make, right?

So in those cases, you really can't differentiate with you because they pretty much predict the same thing.

Um, and, um, you know, and as, as Rowan mentioned, even though it's true, the active inference is kind of motivated from these sort of first principles, um, which is super nice.

Um, the, the thing that you ultimately end up converging on in terms of the decision-making algorithms and active inference start to look pretty similar to the sorts of things that people in machine learning and reinforcement learning have ended up kind of coming to anyway.

They just haven't come to it from a first principles perspective.

larger models um i think that it's really important not to to forget about this aspect of um

trying to not just come up with with you know more architecture and more data but to come up with more interesting frameworks i also think that unlike traditional artificial intelligence or machine learning um it's important to remember that there's this tension between as we mentioned before scalability and um and just the efficiency and biological plausibility we don't always want um a solution a model to just be better and quicker um that isn't necessarily what we're going for we know that in many cases artificial intelligence can do things better and quicker than the human brain but we shouldn't necessarily only care about that sometimes um yeah it's important to get a solution where it it doesn't act more effectively but acts more like um data empirical data that we have gotten from actual human behavior and that it even if it isn't the most efficient sort of architectural algorithmic structure it matches the structure of the brain might implement so i think it's important to remember these factors um in this world where yeah it's fast developing into uh you know big data and massive models um sometimes i find myself uh drawn to just always trying to make active inference faster and always trying to make it more efficient and um oh as long as i have to check myself with that um and and remember what i'm actually interested in so yeah yeah and i guess i guess as a you know concluding thing on my end i mean you know daniel you asked specifically kind of about future directions um you know and and

the kind of thing that we're doing in the lab right now is you know very kind of inspired by what we said you know needs to be done in the um in the in the paper right i mean it's always a little bit of an advertisement for uh in a certain sense for you know here's what we've done and here's what needs to be done and you know that we're trying to kind of follow that ourselves um but um you know and again you know very very much kind of in the context of of trying to also use these frameworks to you know be of some kind of practical benefit to the world right so i mean you know we kind of in tandem right use this as an opportunity to to one you know test theories about you know how brains and minds work but also to um take it as a as a simultaneous opportunity to look for where these sorts of mechanisms may be um you know going wrong a bit and people with different sorts of disorders and whether or not that can um you know give clinicians right like ideas for novel treatment targets right like to target mechanism x versus y um you know as kind of shown by what the differences are and models when we fit them to behavior and say like healthy versus clinical populations um and um and so you know the sorts of things that you know we're doing now on the on the corner more basic science side is is um you know i have another grad student um coping um chow who's not on the call here but you know she's done um some very extensive work now fitting um um something like 50 or so different models um to some decision-making behavior that we have in both a taiwanese sample um where she's from and and an american sample from around here um and you know trying to find right like which model fits best and so these are a range of simpler kind of reinforcement learning models and a range of different active inference models are just parameterized in different ways right so ones that assume sort of like you know static versus dynamic decision noise or static versus dynamic learning or i mean again there's tons of little variants on these things um and so you know she's she's found in in both samples that you know the model that best fits the data um and wins in model comparison is a is a specific um active

inference model that has six parameters i think um that involves a you know dynamic learning or dynamic decision noise um that includes particular sort of forgetting rate function um that's you know not included in kind of the simplest version of active inference um and um you know so showing much more kind of conclusively in two different data sets um you know that um active inference um at least wins in model comparison against some of these other um uh other kind of competing models um that being said i think it's important that even though it wins in model comparison when you actually look at the predictive accuracies um they're very similar you know we're talking about active inference being able to explain like 81.5 of the you know like data versus uh the best reinforcement water reinforcement learning model being able to pick like 80.6 of the behavior you know so i mean these differences are minor right it comes down to being better at predicting just a couple of choices um but sometimes if those choices follow from different sorts of expected dynamics along the way then that can still be meaningful um you know so that's the kind of thing you know that we've been doing on the on the more kind of basic science end but that follows up on you know what we were saying needs to be done um you know on the sort of clinical end still not super clinical as you know we have a study that where that's ongoing right now um that has both an online and a um in-person component where we came up with

five different um five different tasks that um where different models would predict different patterns of behavior um in distinct ways in each of those tasks a couple of them were designed specifically for active inference um others were designed like more in relation to the hgf the hierarchical gaussian filter um and um you know so we're in the process of collecting a lot of data in those where again we'll be able to fit different models and um that one we're also focusing a lot on um into on whether or not we can find models and parameter values in those models that predict individual differences in subjective well-being um as well as um negative affective stuff so we're trying to go for not just predicting like symptoms and negative emotional stuff but also um positive emotional stuff and what makes a difference to decision patterns that are say more consistent with um greater satisfaction with life and subjective well-being and things like that so um you know we're doing that we're also doing some um computational sort of bayesian um modeling of um interoception data so how well people can like detect differences in their um heartbeats or differences in their kind of like respiratory difficulty things like that um and how they're always anxiety disorders so so we're um it's very much kind of a combination of this basic science stuff and uh and trying to move the the usefulness of this thing clinically at the same time so it can be kind of practically helpful to people that's awesome dean and then anyone else and then that will be awesome but where where do you go from here dean um have a shower i don't know i'm not i don't have anything really flat but i appreciate the uh i appreciate the uh i appreciate the explanations and like i said when you start from a place where you take two things and hold them up together at the same time that's that's the part that i uh appreciate here and i think it again reading through all of this um i didn't understand all of the math but um it also i think based on what you uh were able to show i wasn't sort of questioning what what process you went through to draw your conclusions and and state what some of the

limitations are until those tests have have been uh carried out so yeah yeah thank you for the very clear very relevant paper so good luck and you're always welcome back to share more we can follow up and um it'll be an ongoing rolling literature meta-analysis across different systems there will be thousands of fractal sub reviews empirical status in the amygdala on this kind of thing so it's great to hear how you did it this time thanks for having us yeah thanks so much bye bye bye thank you

Thank you.